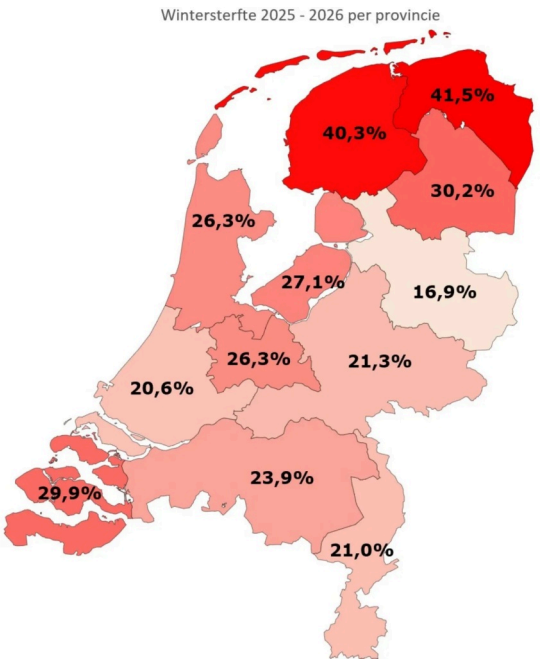
Mortality per province

	A	B	C	D
1		Data		
2	province	Average - death_rate	Median - death_rate	Count - death_rate
3	Drenthe	0,276250793650794	0,166666666666667	25
4	Flevoland	**0,307683982683983**	0,183333333333333	22
5	Friesland, vaste land	0,214285714285714	0	11
6	Gelderland	0,267276737528717	0,0909090909090909	179
7	Groningen	0,287581699346405	0	18
8	Limburg	0,112787546442941	0	54
9	Noord-Brabant	0,220300801168334	0,01	152
10	Noord-Holland	**0,310724098352964**	0,1	97
11	Overijssel	0,156298772391454	0	83
12	Utrecht	0,299181551314587	0,25	85
13	Zeeland	0,314285714285714	0	7
14	Zuid-Holland	**0,339416814571454**	0,222222222222222	97
15	**Total Result**	**0,255776644256314**	**0,0769230769230769**	**830**

Conclusion: there's a mismatch between surveys WUR en Bee Foundation

WUR:



Geographical spread

Total of 830 locations

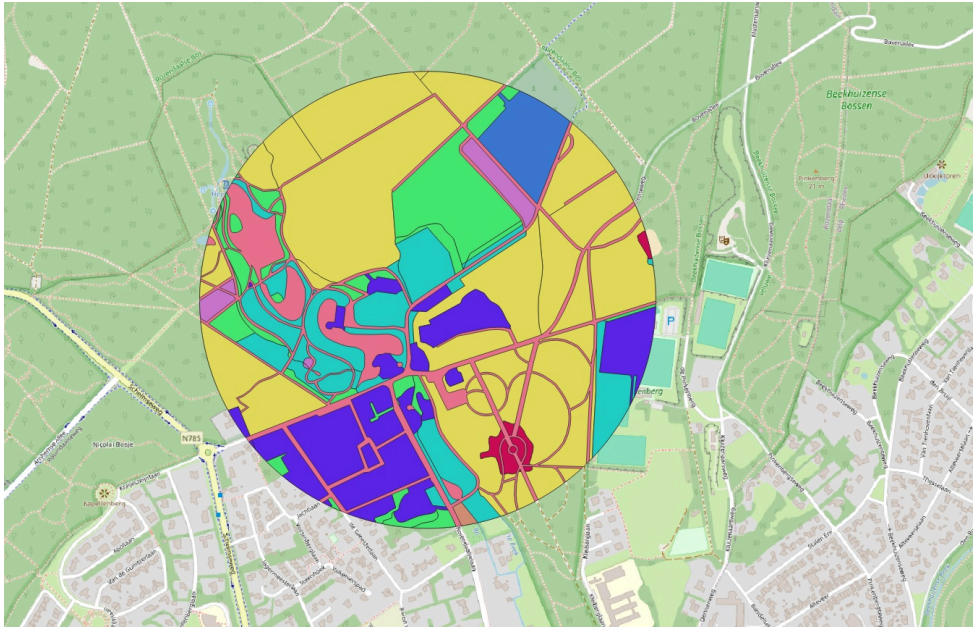
of which Zeeland: 7
Friesland: 11
Groningen: 18

Conclusion: undersampling in certain regions

Note: figure only shows locations with GPS data available (N=426)



Computing a toxicity factor



Using Thomas' tool

Input:
- database Top10NL
- database tabel gewaspercelen

Output:
- per location, all land use types present (crop and other)

Computing a toxicity factor (2)

type	foraging factor	pesticide factor
Aardappelen, bestrijdingsmaatregel AM	1	13
Aardappelen, consumptie	1	16,2
Aardappelen, poot NAK	1	33,7
Aardappelen, poot TBM	1	33,7
Aardappelen, zetmeel	1	13
Aardbeien op stellingen, productie	1	27,2
Aardbeien op stellingen, zaden en opkweekmateriaal	1	27,2
Aardbeien open grond, productie	1	11,2
Aardbeien open grond, vermeerdering	1	11,2
Aardbeien open grond, wachtbed	1	11,2
Aardbeien open grond, zaden en opkweekmateriaal	1	11,2
Aardperen	1	1
Agrarisch natuurmengsel	1	0
…		
Graften	1	0
Granen, overig	0	1
Grasland, blijvend	0	0

Input: CBS data (kg / ha) and some assumptions about where bees forage

$$Tox_k = \frac{\sum_{landtype} Area_{k,landtype} * FF_{landtype} * PF_{landtype}}{\sum_{landtype} Area_{k,landtype} * FF_{landtype}}$$

Computing mortality

```
death_rate =

      (winter_in - winter_uit) / 2025_winter_in
```

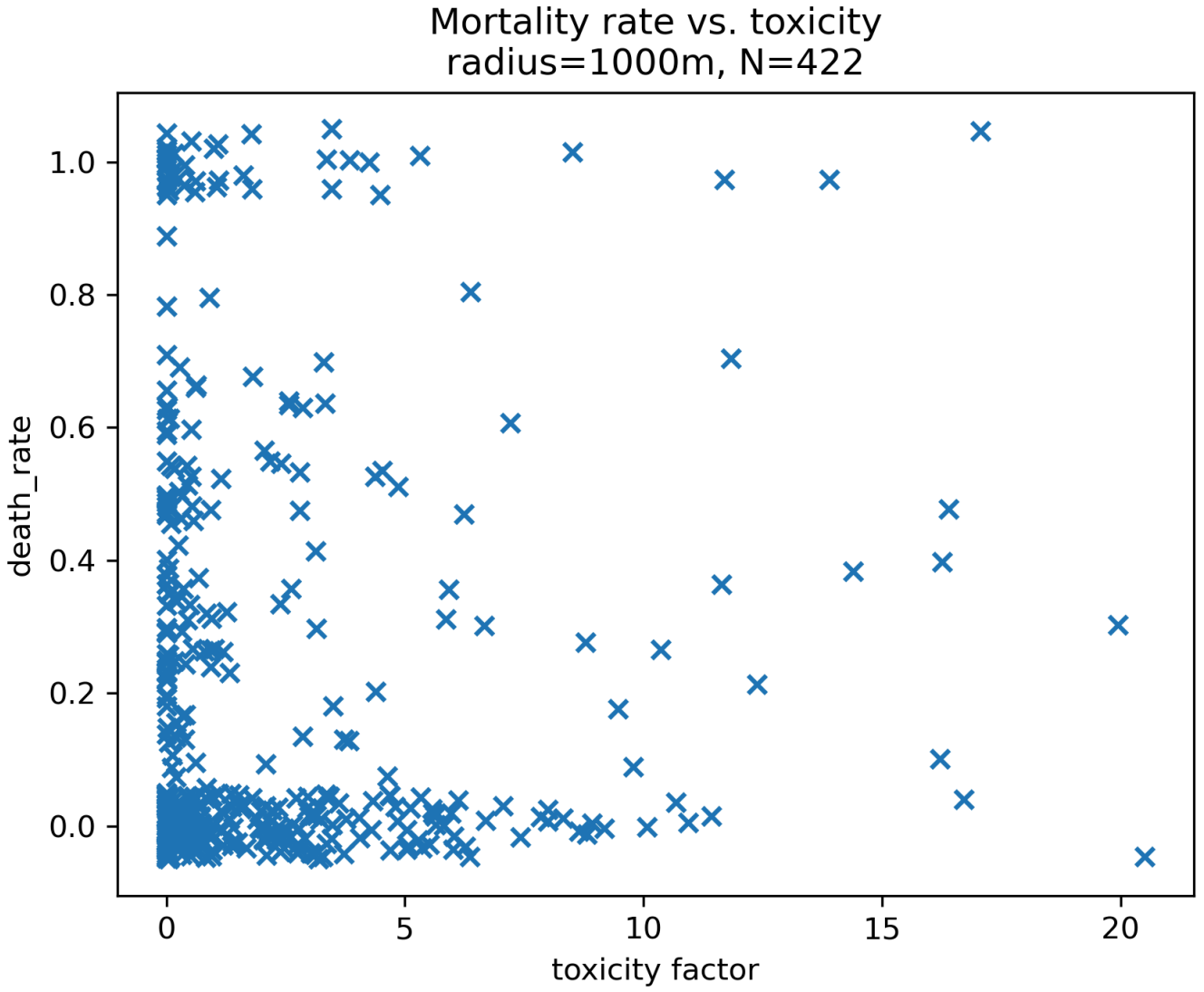
Example

```
[405]: df_mort.iloc[18]
```

```
[405]: id_beekeeper                    31
       province                Noord-Holland
       2025_winter_in                   7
       2025_herfst_dood                 0
       2025_winter_dood                 2
       2026_winter_uit                  5
       winter_in-winter_uit             2
       death_rate                0.285714
       Name: 37, dtype: object
```

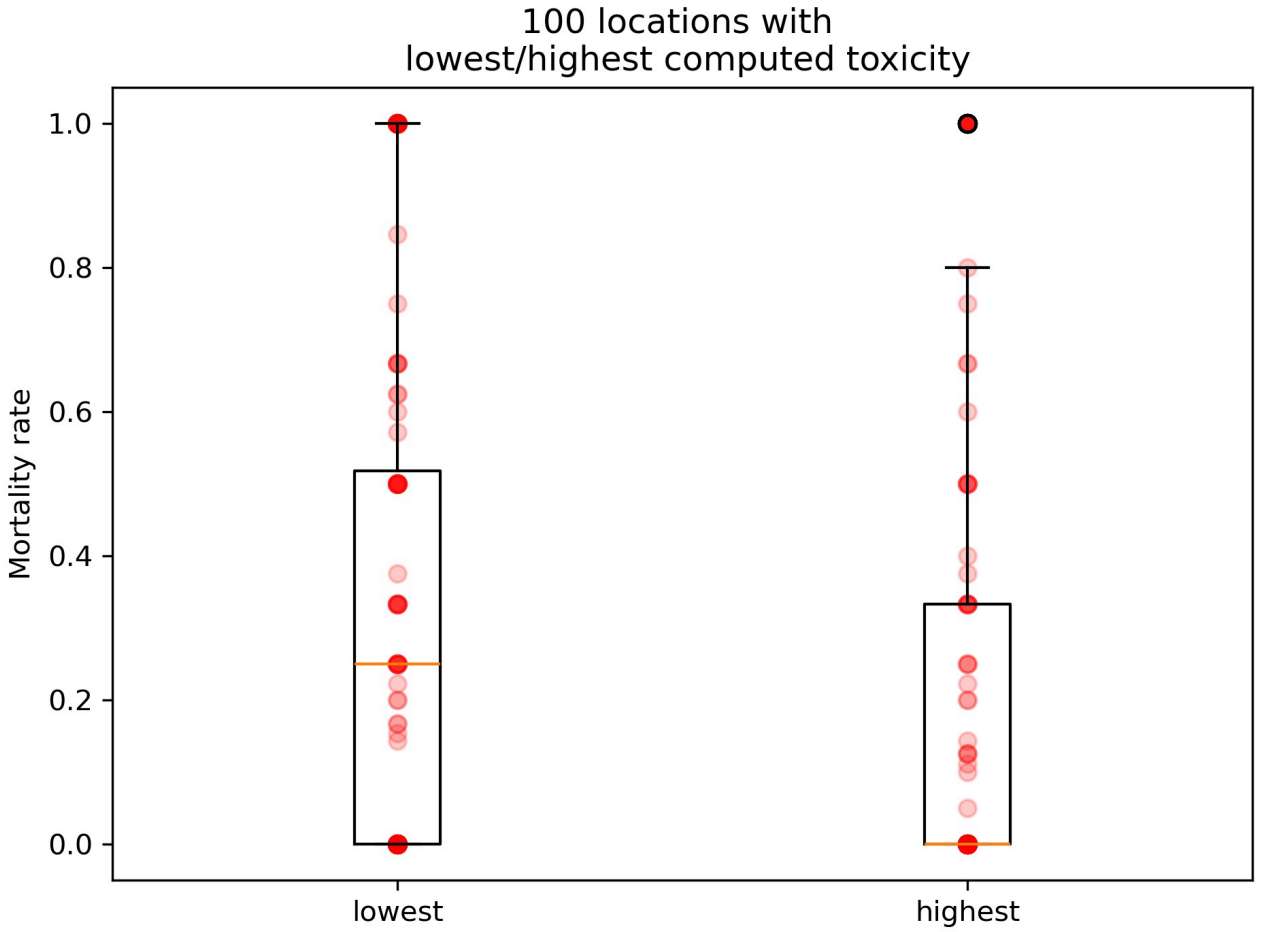
Analysis

- For each location we have computed:
  - toxicity index (Thomas tool, radius 1000 meter)
  - mortality (from survey, $\in[0-1]$)
- Now we can explore whether a relation exists between these, i.e. between pesticide use and winter starvation.



No apparent correlation between the two...

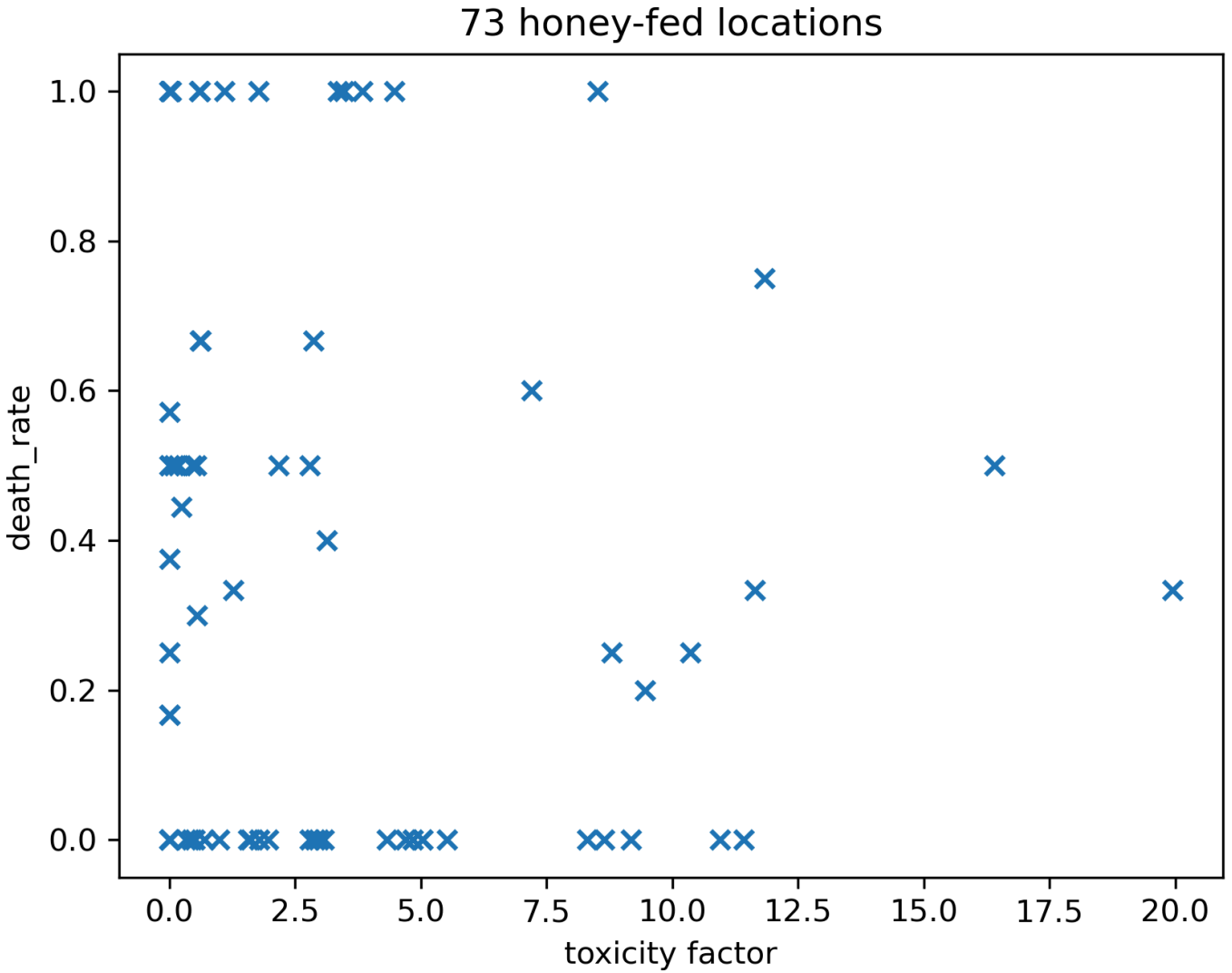
Subset: low and high toxicity



Other determinants: e.g. winter food

	Data	
winter_food	Average - death_rate	Count - death_rate
1 (suikerwater	0,209950289293457	178
2 (combinatie)	0,239772082533851	538
3 (honey)	0,398149549124643	112
(empty)	0,666666666666667	2
Total Result	**0,255776644256314**	**830**

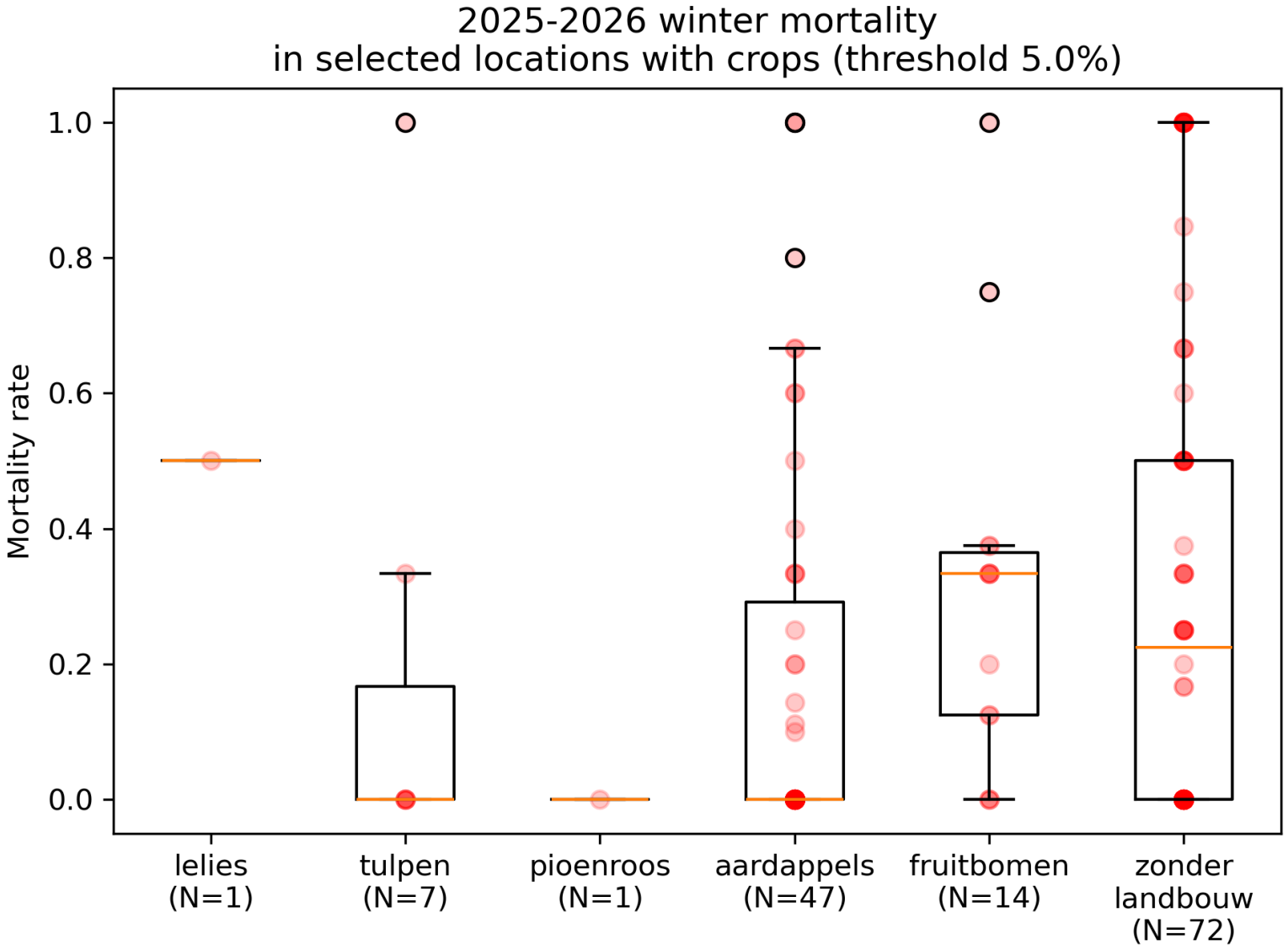
mortality with honey winter food is 40% vs. 21% sugarwater



In subset ‘honey’ winterfood, no apparent connection either

Specific crops

- Next we explore the possibility that some intensive crops will show a clear difference in mortality
- I’m segmenting the locations, selecting them only when certain crops are present (with a threshold of 5% area within the 1000m radius)



too much spread to draw conclusions

Extra: same plot, but with threshold of 1%

